Milestone 2: Line & Connector System Overhaul - Delivery Report

Date: 17 February 2026

Milestone: Milestone 2 - Line & Connector System Overhaul

Status: ✅ Complete

Preview URL: science-canvas-creator.vercel.app

Summary

All Milestone 2 deliverables have been completed and all tests are passing.

Deliverables Completed

- **Consolidated Line Toolbar:** Fragmented line dropdowns replaced with three dedicated toolbar buttons: Straight Line, Curved Line, and Orthogonal Line. Pen and freehand tools unchanged.
- **Smart Connector Point System:** Connection points appear on all canvas objects when a line tool is active. Lines snap-to-attach when dragged near a connection point and snap-to-detach when pulled away. Moving an attached object maintains the line connection. Smart connectors can be toggled in the properties panel.
- **Automatic Pathfinding:** When smart connectors are enabled, orthogonal and bezier lines route around obstacles automatically. Dynamic rerouting triggers when a connected object is moved. Note: rerouting does not trigger when an unconnected object is moved into the line's path — this would require continuous background operations on every canvas object movement, which would degrade performance. Manual override is available via edit mode.
- **Line Edit Mode:** Double-click any placed line to enter edit mode with repositionable anchor and curve handles. Exit by clicking outside the line, pressing Escape, or double-clicking the canvas background.
- **Intelligent Line Type Conversion:** Properties panel shows a line type selector when a line is selected. 2-anchor lines can convert to any type; 3+ anchor lines can only convert to orthogonal or bezier.

- **Customizable Line Endpoints:** Six endpoint options (None, Arrow, Circle, Dot, Diamond, Inhibitor bar) for both start and end, with visual preview in the properties panel and correct orientation across all line types.
- **Line Gradient System:** Gradient controls are available in the properties panel. The requested settings — fade in and reverse direction — are the default values in the gradient section. Gradient is not applied to new lines automatically; users enable it manually, at which point the controls already reflect the correct defaults. Gradients work correctly for straight, curved, orthogonal, and open bezier lines. Closed-loop bezier lines (where the end point connects back to the start) do not currently support gradients — this is a surprisingly complex edge case due to how Fabric.js handles closed path rendering, and can be revisited if required.
- **Advanced Line Styling:** Line weight slider with presets, dash pattern options (solid, dashed, dotted, custom), and corner style options for orthogonal lines (sharp, rounded, beveled).
